

Task 1: Types of hackers

Types of hacker	Description	Real-World example
White Hat Hacker	A security professional who finds vulnerabilities legally and reports them to the owner so they can be fixed.	Security researchers regularly discover and responsibly disclose vulnerabilities in companies such as Google and Microsoft.
Black Hat Hacker	A hacker who accesses systems without permission for theft, damage, fraud, or other malicious purposes.	The WannaCry ransomware attack (2017) is an example of malicious attackers exploiting vulnerable systems and encrypting victims' files.
Grey Hat Hacker	A hacker who may access or test a system without permission but does not necessarily intend to cause harm. They may later report the vulnerability to the organization.	Security researchers have sometimes discovered vulnerabilities in websites without first receiving authorization and subsequently disclosed the issue to the affected company.
Script Kiddie	A person with limited technical knowledge who uses existing hacking tools or scripts created by others rather than developing their own techniques.	News reports frequently describe inexperienced attackers using publicly available ransomware or DDoS tools against websites.
Hacktivist	A hacker or hacking group that conducts	Anonymous has conducted various

	cyber activities to promote a political, social, or ideological cause.	website attacks and online campaigns associated with political and social causes.
State-Sponsored Hacker	A highly skilled attacker supported, funded, or directed by a government, usually for espionage, intelligence gathering, disruption, or strategic purposes.	APT28 (Fancy Bear) has been widely attributed by governments and cybersecurity researchers to Russian military intelligence.

Task 2: Hacker scenarios — Instagram

White Hat Hacker

- A white hat hacker has permission from Instagram to test its security. They might discover a vulnerability that could expose user information and responsibly report it to Meta so that the vulnerability can be fixed.

Black Hat Hacker

- A black hat hacker attempts to gain unauthorized access to Instagram accounts or systems for malicious purposes. For example, they might try to steal credentials or user information and then misuse the compromised accounts.

Grey Hat Hacker

- A grey hat hacker might discover a vulnerability in Instagram without having permission to test the system. They may report the vulnerability afterward, but accessing the system without authorization is still different from an authorized security test.

Task 3: Classify the actions

User Action	Classification	Reason
Reporting a bug to the company	White Hat	The person responsibly reports a security problem so it can be fixed.
Stealing user data	Black Hat	Accessing or stealing data without authorization is malicious and illegal.
Testing your own account for vulnerabilities	White Hat / Authorized Security Testing	You own the account and have permission to test it.
Defacing a website for a political cause	Hacktivist / Black Hat	The political motivation makes it hacktivism, but unauthorized website defacement is generally a malicious/illegal activity.
Participating in a government cyber operation	State-Sponsored	If officially conducted on behalf of a government, it falls under state-sponsored cyber activity.

4. Cricket analogy — White Hat vs Black Hat

Imagine a cricket match where the white hat hacker is like a player or umpire who follows the rules. They are allowed onto the field and use their skills to identify problems, such as noticing that the pitch or equipment has a safety issue, and they report it to the officials. A black hat hacker is like someone who secretly enters the stadium and tampers with the opponent's equipment to gain an unfair advantage. The main difference is that a white

hat works with permission to improve security, while a black hat acts without permission for harmful or unauthorized purposes.